Sohan Jayram Reddy Kadiri

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PROFESSIONAL SUMMARY

Software Development Engineer in Test with experience building automated test frameworks, API and integration suites, and CI quality gates using Python, TypeScript and Java. Skilled in regression testing, root-cause analysis, performance validation and reliability testing across distributed and cloud-based applications.

EDUCATION

Northeastern University Sep 2025 - May 2027
Master of Science, *Computer Science* GPA: 3.9

Siddaganga Institute of Technology Dec 2021 - Jun 2025
Bachelor of Technology, *Artificial Intelligence and Data Science*

WORK EXPERIENCE

Siemens | Software Engineer Intern | Remote Aug 2024 - Jul 2025

- Resolved **9 production defects through log analysis, API tracing and service-level testing**, improving release stability across 2 applications serving 2,500+ users.
- Validated **REST API and AI retrieval workflows across 10,000 benchmark requests**, identifying integration and latency regressions while maintaining 850 ms p95 response latency.
- Led 3 interns through **architecture validation, API integration and regression testing**, reducing escaped production defects by 35% during an 8-week delivery cycle.

Unisys | Software Engineering Intern | Remote Apr 2024 - Jun 2024

- Validated **checkout and marketplace workflows across 8,000+ users and 1,600+ orders**, identifying conversion defects that contributed to a 22% improvement in checkout completion.
- Tested **real-time WebSocket messaging under 350 concurrent connections**, validating sub-150 ms delivery latency and 99.8% message delivery across buyer-seller workflows.
- Performed **cross-device performance and regression validation** using Lighthouse, reducing page load time below 1.2 seconds and increasing mobile conversion by 8%.

PROJECTS

End-to-End QA Automation Framework | Python, Pytest, Playwright, TypeScript, k6, SQLite, axe-core, GitHub Actions

- Built a **multi-layer QA framework spanning manual, SQL, API, UI, performance and accessibility testing**, mapping all 16 requirements to test cases and automating 15 regression checks.
- Automated **API and cross-browser UI validation** using Pytest and Playwright, executing 6 API tests and 36 browser runs across Chromium, Firefox and WebKit.
- Integrated **5 CI quality gates in GitHub Actions** for SQL validation, API tests, UI automation, performance smoke tests and accessibility scans, publishing diagnostic artifacts on every failure.

InspectIQ — Vehicle Inspection Quality & Regression Platform | TypeScript, Python, Playwright, PostgreSQL, AWS, Terraform

- Automated **TypeScript, Python, PostgreSQL integration, browser E2E, visual-regression and mobile test suites** through CI, achieving 92% test coverage and reducing regression-testing time by 40%.
- Validated **S3–SQS–Lambda–Bedrock workflows** using deterministic test providers, schema validation and failure-path testing across asynchronous image-processing pipelines.
- Built a **108-image regression corpus** to validate AI output contracts, detecting 95%+ of schema regressions before deployment and preventing malformed findings from reaching review workflows.

Payment Orchestration Platform — Reliability & Contract Testing | TypeScript, Jest, Supertest, PostgreSQL, RabbitMQ, Testcontainers, GitHub Actions

- Designed **architecture, unit, HTTP and integration test layers** covering service boundaries, payment state transitions, authentication, idempotency and event-contract validation.
- Tested **PostgreSQL row locks, transaction atomicity and RabbitMQ failure paths** with Testcontainers, validating duplicate delivery, reconnect behavior and at-least-once processing guarantees.
- Integrated **contract and reliability tests into CI quality gates**, validating service builds and preventing 90%+ of critical payment-flow regressions from reaching deployable artifacts.

SKILLS

Languages: Python, Java, TypeScript, JavaScript, C++, SQL

Test Automation: : Pytest, Playwright, Jest, Supertest, Catch2, k6, axe-core

Testing: Unit Testing, API Testing, Integration Testing, End-to-End Testing, Regression Testing, Performance Testing, Accessibility Testing, Contract Testing, Fault Injection

Backend & Data: REST APIs, WebSockets, PostgreSQL, Redis, Kafka, RabbitMQ, SQLite

CI/CD & Cloud: GitHub Actions, Docker, AWS, Azure, Terraform, Vercel, CI/CD

Quality Engineering: Test Design, Test Cases, Root-Cause Analysis, Mocking, Test Fixtures, Schema Validation, Test Coverage, Defect Tracking

CERTIFICATIONS & ACHIEVEMENTS

- Open Source - Merged PR #472 into mindsdb/anton, fixing a bug that deleted saved user rules and adding 3 regression tests. Sep 2026
- Publication - "Detection of DDoS Attacks in Fog Computing using Interpretable Quantum-based 2Q Learning Approach" - IEEE Jan 2025